



2013 Half-Yearly Examination

FORM V

MATHEMATICS EXTENSION 1

Wednesday 15th May 2013

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.

Total — 100 Marks

- All questions may be attempted.

Section I – 9 Marks

- Questions 1–9 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 91 Marks

- Questions 10–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and master on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Ten.
- Write your name and master on this question paper and submit it with your answers.

5A: DNW

5B: PKH

5C: RCF

5D: BDD

5E: KWM

5F: FMW

5G: LRP

5H: TCW

Checklist

- SGS booklets — 7 per boy
- Multiple choice answer sheet
- Candidature — 152 boys

Examiner

RCF

SECTION I - Multiple Choice

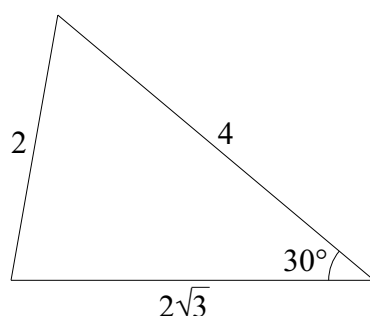
Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

$(2\sqrt{3} + 3\sqrt{2})^2$ simplifies to:

- (A) 30 (B) $30 + 6\sqrt{6}$ (C) $4\sqrt{3} + 9\sqrt{2}$ (D) $30 + 12\sqrt{6}$

QUESTION TWO



The area of the triangle drawn above is:

- (A) $4u^2$ (B) $2\sqrt{3}u^2$ (C) $4\sqrt{3}u^2$ (D) $2\sqrt{6}u^2$

QUESTION THREE

The expression $\cot x \times \sec x$ simplifies to:

- (A) $\sin x$ (B) $\operatorname{cosec} x$ (C) $\cos x$ (D) $\sec x$

QUESTION FOUR

Which of the following is NOT an even function?

- (A) $f(x) = \cos x$ (B) $f(x) = \frac{1}{x}$
(C) $f(x) = x^2 + 3$ (D) $f(x) = -\sqrt{4 - x^2}$

QUESTION FIVE

When $2 \log a + \log b + \log b$ is simplified, the result is:

- (A) $2 \log(ab)$ (B) $\log(2a^2b)$ (C) $\log(a^2 + 2b)$ (D) $\log(2a + 2b)$

Exam continues next page ...

QUESTION SIX

The derivative of $\sqrt[3]{x}$ is:

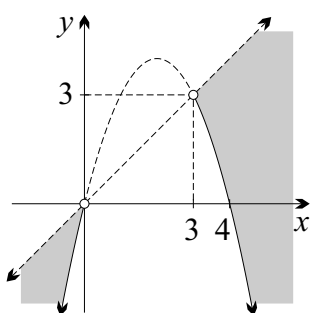
- (A) $3x^{-\frac{2}{3}}$ (B) $\frac{1}{3}x^{-\frac{1}{3}}$ (C) $\frac{1}{3}x^{-\frac{2}{3}}$ (D) $\frac{1}{\sqrt[3]{x}}$

QUESTION SEVEN

The perpendicular distance from the point $(-1, 2)$ to the line $2x - 3y + 4 = 0$ is:

- (A) $\frac{4\sqrt{13}}{13}$ (B) 0 (C) $\frac{8\sqrt{13}}{13}$ (D) $-\frac{4\sqrt{13}}{13}$

QUESTION EIGHT



Which of the following pairs of inequalities best describes the shaded region above?

- (A) $y - x \leq 0$ and $y \geq 4x - x^2$ (B) $y - x < 0$ and $y < 4x - x^2$
 (C) $y - x > 0$ and $y \leq 4x - x^2$ (D) $y - x < 0$ and $y \geq 4x - x^2$

QUESTION NINE

The sum of the first n terms of the geometric sequence $3, -9, 27 \dots$ is given by:

- (A) $\frac{3}{4}(1 - 3^n)$ (B) $\frac{3}{4}\left(1 - (-3)^n\right)$
 (C) $\frac{3}{4}\left((-3)^n - 1\right)$ (D) $\frac{3}{4}(3^n - 1)$

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION TEN (13 marks) Use a separate writing booklet.	Marks
(a) Evaluate:	
(i) $ -4 - 6 $	1
(ii) $\log_3 81$	1
(b) Determine the exact value of $\cos 150^\circ$.	2
(c) Simplify fully:	
(i) $\frac{2x+4}{6x+12}$	1
(ii) $\sqrt{x} \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)$	1
(d) Rationalise the denominator of $\frac{4}{3-\sqrt{2}}$.	2
(e) Consider the points $A(-3, 5)$ and $B(7, -2)$ in the number plane.	
(i) Determine the co-ordinates of the midpoint of AB .	1
(ii) Find the exact length of the interval AB .	1
(iii) Calculate the gradient of the interval AB .	1
(f) (i) Given an arithmetic sequence with first term 4 and common difference 3, what is the tenth term?	1
(ii) Given a geometric sequence with first term 3 and common ratio $\frac{1}{2}$, what is the limiting sum?	1

QUESTION ELEVEN (13 marks) Use a separate writing booklet.

Marks

- (a) (i) State the co-ordinates of the centre of the circle with equation $(x - 3)^2 + (y + 2)^2 = 16$. 1
- (ii) Find the co-ordinates of the vertex of the parabola with equation $y = (x - 4)(x + 2)$. 2
- (b) Differentiate:
- (i) $3x^2 + 4x$ 1
- (ii) x^{-3} 1
- (iii) $3\sqrt{x}$ 1
- (c) Solve:
- (i) $3 - x > 5$ 1
- (ii) $|x + 2| = 7$ 2
- (iii) $x - 2x^2 = 0$ 2
- (iv) $x^2 - 4x - 12 \geq 0$ 2

QUESTION TWELVE (13 marks) Use a separate writing booklet.

Marks

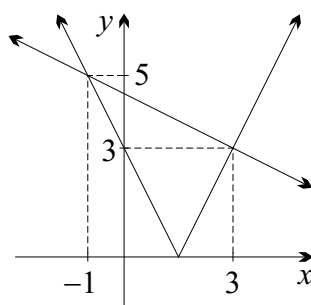
- (a) Given $\sin \theta = \frac{2}{3}$ and $\cos \theta < 0$, find the exact value of $\tan \theta$. 2
- (b) The fourth term of an arithmetic progression is 5 and the seventh term is 23.
- (i) Find the first term and the common difference. 2
- (ii) Find the twentieth term. 1
- (iii) Find the sum of the first twenty terms. 1
- (iv) If the last term is 179, how many terms are there in the AP? 1
- (c) Solve the following trigonometric equations, for $0^\circ \leq \theta \leq 360^\circ$:
- (i) $\cos \theta = \frac{1}{2}$ 1
- (ii) $\sqrt{2} \sin \theta \cos \theta = \sin \theta$ 2
- (d) In $\triangle ABC$, side AB is 6 metres, BC is 11 metres and $\angle BCA$ is 25° . Find the possible values for $\angle BAC$. Give your answers correct to the nearest degree. 3

QUESTION THIRTEEN (13 marks) Use a separate writing booklet.**Marks**(a) Let $f(x) = 2^x$.(i) Write down the equation of the function obtained by translating $y = f(x)$ two units to the right. 1(ii) Write down the equation of the function obtained by reflecting $y = f(x)$ in the x -axis. 1

(b) State the natural domain of:

(i) $y = \frac{1}{2x - 3}$ 1

(ii) $y = -\sqrt{16 - x^2}$ 1

(c) 1

The graph above shows the absolute value function $y = |2x - 3|$ and the straight line $y = \frac{9 - x}{2}$. Use the graph to solve the inequation $|2x - 3| \geq \frac{9 - x}{2}$.

(d) Consider the cubic function $y = 3x - x^3$.(i) Determine the gradient of the tangent to the curve at the point where $x = 2$. 2(ii) Hence find the equation of the normal to the curve at the point where $x = 2$. 2

(e) Differentiate:

(i) $y = \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)$ 2

(ii) $y = \frac{x^4 + 2x^2 + 4}{3x^3}$ 2

QUESTION FOURTEEN (13 marks) Use a separate writing booklet. **Marks**

- (a) State the range of $y = \sec x$. **1**
- (b) Let the points P and Q be $(-2, 5)$ and $(7, -13)$ respectively. Find the co-ordinates of the point R which divides the interval PQ in the ratio $4 : 5$. **2**
- (c) Solve the following trigonometric equations, for $0^\circ \leq \theta \leq 360^\circ$. Approximate your answers to the nearest degree where necessary.
- (i) $3 \cos^2 \theta = 2(1 - \sin \theta)$ **3**
- (ii) $\cos(2\theta - 50^\circ) = \frac{1}{\sqrt{2}}$ **3**
- (d) Solve the inequation $\frac{x}{x-3} \leq 2$. **4**

QUESTION FIFTEEN (13 marks) Use a separate writing booklet. **Marks**

- (a) Consider the function $f(x) = x^2 - 3x$.
- (i) Find $f(x + h)$. **1**
- (ii) Hence differentiate $f(x)$ from first principles. **2**
- (b) Consider the circle $(x - 1)^2 + (y - 3)^2 = 4$ and the straight line $2x - y + k = 0$, where k is a constant.
- (i) Use the perpendicular distance formula to get an expression for the distance from the line to the centre of the circle. **1**
- (ii) Determine the range of values of k for which the line intersects the circle. **2**
- (c) Prove the following trigonometric identities:
- (i) $\cot \theta + \tan \theta = \sec \theta \operatorname{cosec} \theta$ **2**
- (ii) $\frac{\operatorname{cosec} \alpha}{\operatorname{cosec} \alpha - \sin \alpha} = \sec^2 \alpha$ **2**
- (d) In a certain geometric sequence, the sum of the first two terms is 9 and the sum of the first three terms is 21. Find the possible values of the common ratio. **3**

QUESTION SIXTEEN (13 marks) Use a separate writing booklet.**Marks**

(a) Consider the rational function $f(x) = \frac{x^2 + 2x - 3}{2x^2 - x - 3}$.

(i) Find the x -intercepts and y -intercept of the graph of $y = f(x)$.

2

(ii) Find the equations of any vertical asymptotes, showing clear working.

1

(iii) Find the equation of the horizontal asymptote, showing clear working.

1

(iv) Find the x value where the graph crosses the horizontal asymptote.

1

(v) Carefully sketch the function, showing these features.

1

(b) Find x given $\sum_{n=1}^{2x} \left(\sqrt[4]{n} + \sqrt[4]{n+1} \right) \left(\sqrt[4]{n+1} - \sqrt[4]{n} \right) = 2$.

3

(c) Inside a large semicircle with centre O and diameter AOB of length $2R$, a smaller semicircle is drawn with diameter AO and centre P . A small circle is drawn with centre Q and radius r which is tangent to the large semicircle at T and the small semicircle at S . Let $\angle POQ = \theta$.

(i) Carefully draw a diagram of the situation described.

1

(ii) What can be said about the points O , Q and T ? Justify your answer.

1

(iii) Use the cosine rule to show that

2

$$\frac{r}{R} = \frac{1 - \cos \theta}{3 - \cos \theta}.$$

————— End of Section II —————

END OF EXAMINATION



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

NAME:

CLASS: MASTER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Extension One 2013 Half Yearly

Question One

$$(2\sqrt{3} + 3\sqrt{2})(2\sqrt{3} + 3\sqrt{2}) = 12 + 18 + 12\sqrt{6} = 30 + 12\sqrt{6} \quad (D) \checkmark$$

Question Two

$$A = \frac{1}{2}ab \sin C = \frac{1}{2} \times 4 \times 2\sqrt{3} \times \sin 30^\circ = 4\sqrt{3} \times \frac{1}{2} = 2\sqrt{3} \text{ u}^2 \quad (B) \checkmark$$

Question Three

$$\cot x \sec x = \frac{\cos x}{\sin x} \times \frac{1}{\cos x} = \frac{1}{\sin x} = \csc x \quad (B) \checkmark$$

Question Four

$$f(x) = \frac{1}{x} \text{ has ODD symmetry} \quad (B) \checkmark$$

Question Five

$$2\log a + \log b + \log b = \log a^2 + \log b^2 = \log a^2 b^2 = 2\log ab \quad (A) \checkmark$$

Question Six

$$y = x^{\frac{1}{3}} \quad \frac{dy}{dx} = \frac{1}{3}x^{-\frac{2}{3}} \quad (C) \checkmark$$

Question Seven

$$2x - 3y + 4 = 0 \quad (-1, 2) \quad \rho = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|(-2) + (-6) + 4|}{\sqrt{4 + 9}} = \frac{4}{\sqrt{13}} = \frac{4\sqrt{13}}{13} \quad (A) \checkmark$$

Question 8

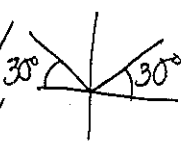
$$y < x \text{ ie } y - x < 0 \text{ and } y > 4x - x^2 \quad (D) \checkmark$$

Question 9

$$GP \quad a = 3 \quad r = (-3) \quad S_n = \frac{a(1-r^n)}{1-r} = \frac{3(1-(-3)^n)}{1-(-3)} = \frac{3}{4}(1-(-3)^n) \quad (B) \checkmark$$

Question 10

$$(a) (i) |-4| - |-6| = 4 - 6 = -2 \quad (ii) \log_3 81 = 4 \quad \checkmark$$

$$(b) \cos 150^\circ = -\cos 30^\circ = -\frac{\sqrt{3}}{2} \quad \checkmark$$


$$(c) (i) \frac{2(x+2)}{6(x+2)} = \frac{1}{3} \quad (ii) \sqrt{x}(\sqrt{x} - \frac{1}{\sqrt{x}}) = x - 1 \quad \checkmark$$

$$(d) \frac{4}{3-\sqrt{2}} \times \frac{(3+\sqrt{2})}{(3+\sqrt{2})} = \frac{12+4\sqrt{2}}{9-2} = \frac{12+4\sqrt{2}}{7} \quad \checkmark \text{ or } \frac{4(3+\sqrt{2})}{7}$$

$$(e) A(-3, 5) \quad B(7, -2)$$

$$(i) M\left(\frac{-3+7}{2}, \frac{5+(-2)}{2}\right) = \left(2, \frac{3}{2}\right) \quad (ii) m_{AB} = \frac{5-(-2)}{(-3)-7} = \frac{7}{-10} = -\frac{7}{10} \quad \checkmark$$

$$(ii) AB = \sqrt{10^2 + 7^2} = \sqrt{149} \text{ u} \quad \checkmark$$

(c) VNT a-7 a-5

$U_{10} = a + 9d$
 $= 4 + 27$
 $= 31$ / tenth term is 31.

(ii) GP $a=3$ $r=\frac{1}{2}$

$S_{\infty} = \frac{a}{1-r} = \frac{3}{1-\frac{1}{2}} = 6$ / limiting sum is 6.

Question Eleven

(a)(i) $(x-3)^2 + (y+2)^2 = 16$
 Circle
 Centre $(3, -2)$ ✓

(ii) $Y = (x-4)(x+2)$
 x-intercepts are $(4, 0)$ and $(-2, 0)$
 x co-ord of vertex is $x=1$ ✓
 y co-ord $y = -3 \times 3 = -9$ ✓
 Vertex $(1, -9)$

b)(i) $y = 3x^2 + 4x$
 $\frac{dy}{dx} = 6x + 4$ ✓

(ii) $y = 3x^{\frac{1}{2}}$
 $\frac{dy}{dx} = \frac{3}{2}x^{-\frac{1}{2}}$ ✓

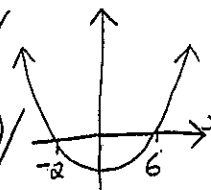
(i) $y = x^{-3}$
 $\frac{dy}{dx} = -3x^{-4}$ ✓

c)(i) $3-x > 5$
 $(+x) \quad (+x)$
 $3 > x+5$
 $(-5) \quad (-5)$
 $(-2) > x$ ✓

(ii) $|x+2| = 7$
 $x+2=7$ / or $x+2=-7$
 $x=5$ / or $x=-9$ ✓

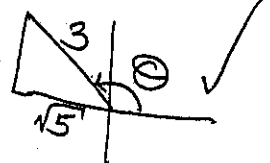
(iii) $x-2x^2=0$
 $x(1-2x)=0$
 $x=0$ / or $x=\frac{1}{2}$ ✓

(iv) $x^2 - 4x - 12 \geq 0$
 $(x-6)(x+2) \geq 0$ ✓
 $x \geq 6$ or $x \leq -2$ ✓



Exercises 1.1

a) $\sin \theta = \frac{2}{3}$ $\cos \theta < 0 \Rightarrow 2^{\text{nd}} \text{ quad}$ ✓
 $\tan \theta = \left(-\frac{2}{\sqrt{5}}\right)$ ✓



b) AP
 $a+3d=5$ ①
 $a+6d=23$ ②

First term is (-13)
 Common difference is 6.

② - ① $3d=18$
 $d=6$ ✓
 Sub into ① $a+18=5$
 $a=(-13)$ ✓

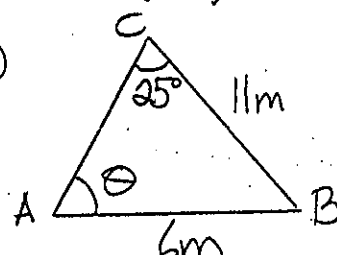
(ii) $t_{20} = a + 19d$
 $= (-13) + 19 \times 6$
 $= (-13) + 114$
 $= 101$ ✓

(iii) $S_n = \frac{n}{2}(2a + (n-1)d)$ or $S_n = \frac{n}{2}(a+l)$
 $= 10(-26 + 19 \times 6)$
 $= 10(114 - 26)$
 $= 880$ ✓
 $= 10(-13 + 101)$
 $= 880$

(iv) $t_n = 179 = a + (n-1)d$
 $(-13) + 6(n-1) = 179$
 $6n - 19 = 179$
 $6n = 198$
 $n = 33$ ✓
 There are 33 terms in AP.

2(i) $\cos \theta = \frac{1}{2}$
 $\theta = 60^\circ, 300^\circ$ ✓

(ii) $\sqrt{2} \sin \theta \cos \theta = \sin \theta$
 $\sin \theta (\sqrt{2} \cos \theta - 1) = 0$ ✓
 $\sin \theta = 0$ or $\cos \theta = \frac{1}{\sqrt{2}}$
 $\theta = 0^\circ, 180^\circ, 360^\circ$ or $\theta = 45^\circ, 315^\circ$ ✓

d)  Let $\angle BAC = \theta$
 $\frac{\sin \theta}{11} = \frac{\sin 25^\circ}{6}$ ✓
 $\sin \theta = \frac{11 \sin 25^\circ}{6}$
 $\approx 0.7748 \dots$
 $\theta \approx 51^\circ$ or 129° ✓
 Note: Ambiguous Case

Question Thirteen

a) (i) $f(x) = 2^{x-2}$ ✓ (ii) $g(x) = -2^x$ ✓

b) (i) $y = \frac{1}{2x-3}$ $\therefore 2x-3 \neq 0$ ✓
 $x \neq \frac{3}{2}$

Domain $\{x: x \in \mathbb{R}, x \neq \frac{3}{2}\}$

(ii) $y = -\sqrt{16-x^2}$ $\therefore 16-x^2 \geq 0$
 (or recognise lower semicircle) $(4-x)(4+x) \geq 0$ ✓
 $(-4) \leq x \leq 4$
 Domain $\{x: x \in \mathbb{R}, -4 \leq x \leq 4\}$



(c) Absolute Value above straight line
 $x \geq 3$ or $x \leq (-1)$ ✓

(d) (i) $y = 3x - x^3$
 $\frac{dy}{dx} = 3 - 3x^2$ ✓

$\left(\frac{dy}{dx}\right)_{x=2} = 3 - 3(2)^2 = (-9)$ / Gradient of tangent is (-9)

(ii) $m_{\text{normal}} = \frac{1}{9}$ Eqn of normal $y - (-2) = \frac{1}{9}(x - 2)$
 $y + 2 = \frac{1}{9}(x - 2)$
 $9y + 18 = x - 2$
 $0 = x - 9y - 20$ ✓

(e) (i) $y = \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)$
 $= x - \frac{1}{x}$
 $= x - x^{-1}$ ✓

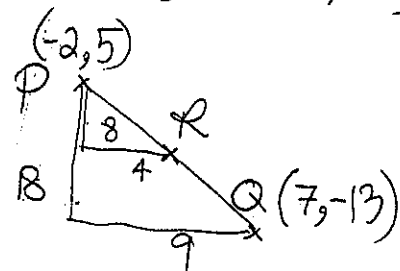
$\frac{dy}{dx} = 1 + x^{-2}$
 $= 1 + \frac{1}{x^2}$ ✓
 $= \frac{x^2 + 1}{x^2}$

(ii) $y = \frac{x^4 + 2x^2 + 4}{3x^3}$
 $= \frac{1}{3}x + \frac{2}{3}x^{-1} + \frac{4}{3}x^{-3}$ ✓
 $\frac{dy}{dx} = \frac{1}{3} - \frac{2}{3}x^{-2} - \frac{4}{3}x^{-4}$
 $= \frac{1}{3} - \frac{2}{3x^2} - \frac{4}{3x^4}$ ✓
 $= \frac{x^4 - 2x^2 - 4}{3x^4}$

Question Fourteen

a) $y = \sec x$ $y \geq 1$ or $y \leq (-1)$ ✓

" P(-2, 5) Q(7, -13) in ratio 4:5



Point R

$$x = \frac{5(-2) + 4 \times 7}{4+5}$$

$$= \frac{18}{9}$$

$$= 2 \quad \checkmark$$

R(2, -3)

$$y = \frac{5 \times 5 + 4 \times (-13)}{4+5}$$

$$= \frac{25-52}{9}$$

$$= -3 \quad \checkmark$$

(d) $\frac{x}{x-3} \leq 2, (x \neq 3)$

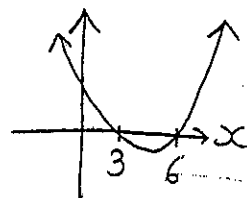
$$(x-3)^2 \checkmark$$

$$x(x-3) \leq 2(x-3)^2$$

$$0 \leq 2(x-3)^2 - x(x-3)$$

$$0 \leq (x-3)[2x-6-x] \checkmark$$

$$0 \leq (x-3)(x-6) \checkmark$$



$\therefore$ Solns $x < 3$ / or $x \geq 6$ /
(NB $x \neq 3$)

c) i) $3\cos^2 \theta = 2(1 - \sin \theta)$

$$3\cos^2 \theta + 2\sin \theta - 2 = 0$$

$$3(1 - \sin^2 \theta) + 2\sin \theta - 2 = 0$$

$$-3\sin^2 \theta + 2\sin \theta + 1 = 0 \quad \checkmark$$

$$3\sin^2 \theta - 2\sin \theta - 1 = 0 \quad \checkmark$$

$$(3\sin \theta + 1)(\sin \theta - 1) = 0$$

$$\therefore \sin \theta = (-\frac{1}{3}) \text{ or } \sin \theta = 1 \quad \checkmark$$

$$\theta = 199^\circ, 341^\circ \quad \theta = 90^\circ \quad \checkmark$$

ii) $\cos(2\theta - 50^\circ) = \frac{1}{\sqrt{2}}$ $0^\circ \leq \theta < 360^\circ$

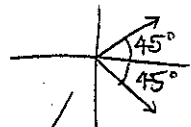
$$0^\circ \leq 2\theta < 720^\circ$$

$$2\theta - 50^\circ = -45^\circ, 45^\circ, 315^\circ, -50^\circ \leq 2\theta - 50^\circ < 670^\circ \quad \checkmark$$

$$405^\circ, 675^\circ$$

$$2\theta = 5^\circ, 95^\circ, 365^\circ, 455^\circ$$

$$\theta = 2\frac{1}{2}^\circ, 47\frac{1}{2}^\circ, 182\frac{1}{2}^\circ, 227\frac{1}{2}^\circ \quad \checkmark$$



Question Fifteen

(a) $f(x) = x^2 - 3x$

i) $f(x+h) = (x+h)^2 - 3(x+h)$

$$= x^2 + 2xh + h^2 - 3x - 3h \quad \checkmark$$

ii) $f'(x) = \lim_{h \rightarrow 0} \left\{ \frac{f(x+h) - f(x)}{(x+h) - x} \right\}$

$$= \lim_{h \rightarrow 0} \left\{ \frac{x^2 + 2xh + h^2 - 3x - 3h - (x^2 - 3x)}{h} \right\} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \left\{ \frac{2xh + h^2 - 3h}{h} \right\}$$

$$= \lim_{h \rightarrow 0} \{ 2x - 3 + h \} \quad \checkmark$$

$$= 2x - 3$$

2) $(x-1)+(y-3) = -1$
 Centre $(1,3)$ $r=2$ $y=2x+k$
 $\therefore 2x-y+k=0$
 $a=2, b=(-1), c=k$

$$p = \frac{|2 \times 1 + 3 \times (-1) + k|}{\sqrt{4+1}}$$

$$= \frac{|k-1|}{\sqrt{5}}$$

For intersection or tangency $p \leq r$
 $\frac{|k-1|}{\sqrt{5}} \leq 2$ ✓
 $|k-1| \leq 2\sqrt{5}$

$-2\sqrt{5} \leq k-1 \leq 2\sqrt{5}$
 $1-2\sqrt{5} \leq k \leq 2\sqrt{5}+1$ ✓

2) (i) $\cot \theta + \tan \theta = \sec \theta \operatorname{cosec} \theta$

LHS = $\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta}$
 $= \frac{\cos^2 \theta}{\sin \theta \cos \theta} + \frac{\sin^2 \theta}{\sin \theta \cos \theta}$ ✓
 $= \frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta}$
 $= \frac{1}{\sin \theta \cos \theta}$ ✓
 $= \operatorname{cosec} \theta \sec \theta$
 $= \text{RHS}$

1) $\frac{\operatorname{cosec} \alpha}{\operatorname{cosec} \alpha - \sin \alpha} = \sec^2 \alpha$

LHS = $\frac{\frac{1}{\sin \alpha}}{\frac{1}{\sin \alpha} - \sin \alpha} \times \frac{(\sin \alpha)}{(\sin \alpha)}$
 $= \frac{1}{1 - \sin^2 \alpha}$ ✓
 $= \frac{1}{\cos^2 \alpha}$ ✓
 $= \sec^2 \alpha$
 $= \text{RHS}$

(d) $a+ar = 9$ ①
 $a+ar+ar^2 = 21$ ②

②-① $ar^2 = 12$ ✓ ③

③ ÷ ① $\frac{ar^2}{a+ar} = \frac{12}{9}$

$\frac{r^2}{1+r} = \frac{4}{3}$

$3r^2 = 4r + 4$ ✓

$3r^2 - 4r - 4 = 0$

$(3r+2)(r-2) = 0$

$r = (-\frac{2}{3})$ OR $r = 2$ ✓

Check
 if $r = (-\frac{2}{3})$

① $a - 2a = 9$

$\frac{a}{3} = 9$

$a = 27$

GP $27, -18, 12, \dots$ ✓ (works)

if $r = 2$

③ $\frac{a}{3} = 12$

$a = 36$

GP $= 36, 72, 144, \dots$ ✓ (works)

$$(a) f(x) = \frac{x^2 + 2x - 3}{2x^2 - x - 3}$$

$$= \frac{(x+3)(x-1)}{(2x-3)(x+1)}$$

(i) x -into $f(x) = 0$ $x = -3$ or $x = 1$ ✓
 y -into $x = 0$ ie $(-3, 0), (1, 0)$ ✓
 $f(0) = -\frac{3}{3} = 1$ ✓
 ie $(0, 1)$ ✓

$$(ii) f(x) = \frac{x^2 + 2x - 3}{2x^2 - x - 3} \quad \begin{matrix} \div x^2 \\ \div x^2 \end{matrix}$$

$$= \frac{1 + \frac{2}{x} - \frac{3}{x^2}}{2 - \frac{1}{x} - \frac{3}{x^2}}$$

Horizontal Asymptote
 $y = \frac{1}{2}$

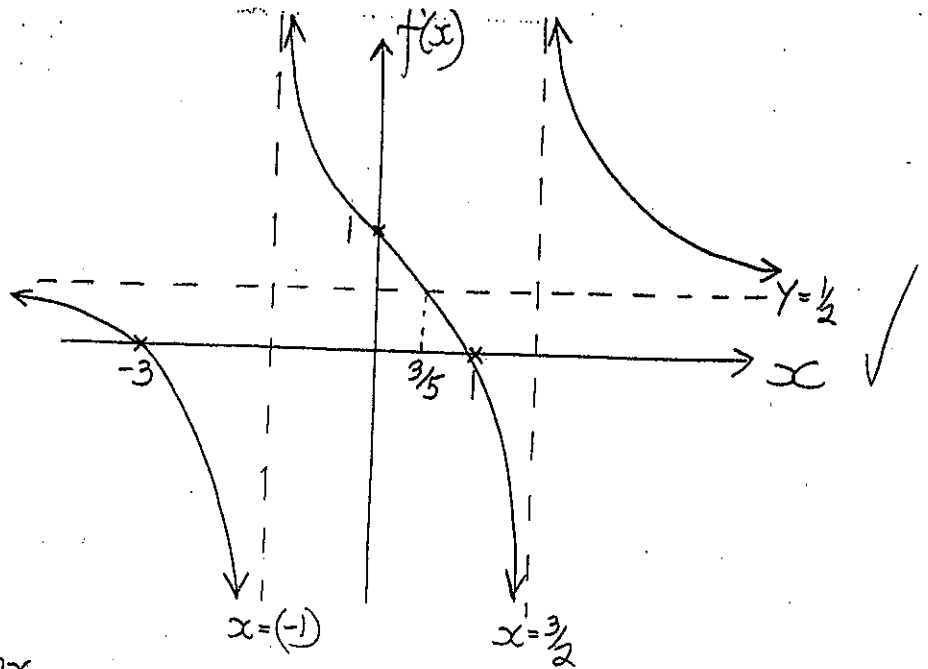
(ii) Vertical Asymptotes where denominator is zero and numerator non-zero
ie $x = -1$ or $x = \frac{3}{2}$ Both

(iv) Graph crosses horizontal asymptote where $y = \frac{1}{2}$
 $\therefore k = x^2, x = 3$

$$\therefore \frac{1}{2} = \frac{x^2 + 2x - 3}{2x^2 - x - 3}$$

$$2x^2 - x - 3 = 2x^2 + 4x - 6$$

$$\begin{aligned} 3 &= 5x \\ \frac{3}{5} &= x \quad \checkmark \text{ ie } (\frac{3}{5}, \frac{1}{2}) \end{aligned}$$



(b) $\sum_{n=1}^{2x} (\sqrt{n+1} + \sqrt{n})(\sqrt{n+1} - \sqrt{n}) = 2$ Diff of 2 sqs

$$\sum_{n=1}^{200} (\sqrt{n+1} - \sqrt{n}) = 2$$

$$\begin{aligned} & (\sqrt{2}-\sqrt{1}) + (\sqrt{3}-\sqrt{2}) + \dots + (\sqrt{2x}-\sqrt{2x-1}) + (\sqrt{2x+1}-\sqrt{2x}) \\ & \sqrt{2x+1}-\sqrt{1}=2 \quad \checkmark \end{aligned}$$

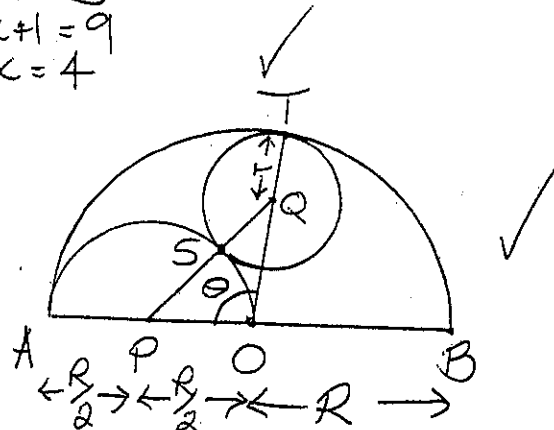
$$\sqrt{2x+1} - \sqrt{1} = 2$$

$$\sqrt{2x+1} = 3$$

$$2x + 1 = 0$$

$$x = 4$$

(c) (i)



(ii) Points O, Q and T are collinear since T is point of contact of common tangent of circles centred at O and Q . Similarly P, S and Q are collinear with common tangent at S . ✓

(iii) Cosine Rule in ΔPOQ . $PO = \frac{R}{2}$

$$PQ^2 = OP^2 + OQ^2 - 2 \times OP \times OQ \times \cos \theta \quad OQ = (R-r)$$

$$\left(\frac{R+r}{2}\right)^2 = \left(\frac{R}{2}\right)^2 + (R-r)^2 - 2\left(\frac{R}{2}\right)(R-r)\cos \theta \quad / \quad PQ = \left(\frac{R+r}{2}\right)$$

$$\cancel{\frac{R^2}{4}} + Rr + \cancel{R^2} = \cancel{\frac{R^2}{4}} + R^2 - 2Rr + R^2 - R(R-r)\cos \theta$$

$$R(R-r)\cos \theta = R^2 - 3Rr$$

$$(\div R^2) \quad (\div R^2)$$

$$\left(\frac{R^2 - Rr}{R^2}\right)\cos \theta = \frac{R^2 - 3Rr}{R^2}$$

$$\text{Let } \frac{R}{r} = k. \quad \therefore (k^2 - k)\cos \theta = k^2 - 3k$$

$$(\div k) \quad (\div k)$$

$$(k-1)\cos \theta = k-3$$

$$k\cos \theta - k = \cos \theta - 3$$

$$k(\cos \theta - 1) = \cos \theta - 3$$

$$k = \frac{\cos \theta - 3}{\cos \theta - 1}$$

$$\frac{R}{r} = \frac{\cos \theta - 3}{\cos \theta - 1} \quad (x-1)$$

$$\frac{R}{r} = \frac{3 - \cos \theta}{1 - \cos \theta}$$

$$\frac{r}{R} = \frac{1 - \cos \theta}{3 - \cos \theta}$$